Francesco **Iacovelli**

Contact

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Languages

Italian (Native), English (Fluent), French (Intermediate).

Programming Languages

Advanced level: C, C++, R, Python, MATLAB, Microsoft Office, LaTeX. Intermediate level: Visual Basic, Arduino, Wolfram Mathematica, FORTRAN, CSS.

Skills and Knowledge

Cosmology and Gravity, Simulations and Data Analysis, Responsibility, Team working, Leadership, Flexibility, Teaching.

Collaboration Membership

Einstein Telescope
Cosmic Explorer
LGWA
LISA

Hobbies

Percussion instruments, photography, tennis, hiking.

Education

2020–2024 **Ph.D.** in Theoretical Physics — University of Geneva, Geneva, Switzerland

Gravitational waves, Cosmology, Modified Gravity and Computing. On November 1, 2020, I moved to Geneva to complete my education, under the supervision of Prof. M. Maggiore. The core focus of my current research activity is the theoretical study of the new and promising field of gravitational waves. To be specific I am focusing on the possibility of using them for cosmological purposes and to test modifications of the theory of General Relativity. Also, being member of the Observational Science Board, I am working on forecasting the capabilities of the Einstein Telescope. Moreover, I am an assistant, thus I am becoming much more proficient in teaching and understanding the other's various points of view. In 2022 I was awarded the Istituto Svizzero "Milano Calling" fellowship, which allowed me to visit the University of Milano-Bicocca from October 2022 to April 2023. During this experience I worked with the group of Prof. D. Gerosa, mainly focusing on Astrophysical aspects related to gravitational waves and data analysis.

2018–2020 **Master's Degree** in Astronomy and Astrophysics — Sapienza University, Rome, Italy

General Relativity, Physical and Theoretical Cosmology, Stellar and High Energy Astrophysics, Relativistic Quantum Mechanics. Different topics linked to Astrophysics and High Energy Physics examined in-depth with particular focus on theoretical aspects of Gravity and Cosmology. In February 2019 I became part of the *Excellence Programme* and researched material linked to Inflation and Effective Field Theory under the supervision of Prof. F. Riccioni. I graduated with honors on October 1, 2020, writing my thesis under the supervision of Prof. M. Maggiore and Prof. A. Melchiorri about cosmological model testing via gravitational waves and galaxy clustering observation.

2015–2018 **Bachelor's Degree** in Physics — Sapienza University, Rome, Italy

Classical, Statistical and Quantum Mechanics, Electromagnetism and Optics, Computational Physics, Statistics, Real, Complex and Functional Analysis. I learned the foundations of both theoretical and experimental physics, as well as the mathematics behind them. I graduated with honors on October 22, 2018, my thesis was about dark matter evidences from gravitational microlensing, the supervisor was Prof. P. de Bernardis.

Professional Appointments

2024–current **Assistant Research Scientist, Postdoctoral Fellowship** — Johns Hopkins University, Baltimore, US

Gravitational waves, Astrophysics and Cosmology. In October 2024 I joined the Astrophysics group at Johns Hopkins University as an Assistant Research Scientist under a Miller Postdoctoral Fellowship. Having the possibility of conducting my research independently, I plan to focus on data analysis for future gravitational wave instruments, with a principal focus on extracting astrophysical and cosmological information from gravitational–wave sources, working in close contact with the group of Prof. E. Berti.

Experience

2021–2024 **Teaching assistant** — University of Geneva, Geneva, Switzerland

Since arriving in Geneva, I have been a teaching assistant for Thermodynamics, Mathematical Methods for Physics, and Electrodynamics II courses. This greatly improved my teaching and organizational skills and allowed me to understand better how to share my knowledge arousing interest in the students.

2022–2024 **Maturity exam jury member** Geneva, Switzerland

In 2022 (Collège de Saussure), 2023 (Collège Rousseau, Collège Sismondi, CEC André-Chavanne) and 2024 (Collège de Saussure, Collège Madame de Staël) I have been a jury member during the Swiss maturity exams, for the subjects of "Physics and Applications of Mathematics" and "Mathematics". As such, I participated in the correction of the written tests and oral examinations. This improved my ability in testing and evaluating students knowledge.

2019 **Collaboration Scholarship** Sapienza University, Rome, Italy

Organizational, teaching and problem solving skills. I set up instrumentation both for Mechanics, Thermodynamics and Electromagnetism Laboratory experimental classes and helped younger students during data taking and report writing.

Publications

Publications: **24** papers published in major peer–reviewed journals, **4** papers in submission stage, **3** public codes, **5** proceedings.

Total number of citations: 1740 [iNSPIRE] – 1466 [NASA/ADS] – 1634 [Scholar]

h-index: 15

Alphabetical order is the standard publication policy in Theoretical Cosmology, regardless of the actual contribution of the authors, and the group where I am working during my Ph.D. usually adopts such practice. I will therefore add a brief summary of my contribution to each paper I am not the first author of.

Links to publication list: iNSPIRE, NASA/ADS, Scholar

Published papers

[24] S. Yi, **F. Iacovelli**, S. Marsat, D. Wadekar, E. Berti. "Systematic biases in parameter estimation on LISA binaries: The effect of excluding higher harmonics for non-spinning binaries". In *Phys. Rev. D* 112, 124063 (2025). DOI: 10.1103/3gs3-gmb4. arXiv: 2502.12237 [gr-qc].
I contributed to the conceptual development of the paper, the writing and the production and validation of the novel analysis tools.

[23] A. Abac, R. Abramo, et al. (incl. **F. Iacovelli**). "The Science of the Einstein Telescope". To appear in *JCAP* 03 (2026) 081. DOI: 10.1088/1475-7516/2026/03/081. arXiv: 2503.12263 [gr-qc].
I took part in the writing and production of the results for several sections, in particular: Cosmology with ET, Multi-messenger observations in the ET era, Synergies of ET with other gravitational-wave observatories, and Tools for assessing the scientific potentials of detector configurations. For the latter I had a leading role.

[22] E. Belgacem, **F. Iacovelli**, M. Maggiore, M. Mancarella, N. Muttoni. "Confusion noise from astrophysical backgrounds at third-generation gravitational-wave detector networks". In *Phys. Rev. D* 112, 083007 (2025). DOI: 10.1103/mr25-ftqd. arXiv: 2411.04029 [gr-qc].
I contributed to the theoretical development of the paper, the writing and produced most of the numerical results.

[21] V. De Luca, L. Del Grosso, **F. Iacovelli**, A. Maselli, E. Berti. "Tainted Love: Systematic biases from ignoring environmental tidal effects in gravitational wave observations". In *Phys. Rev. D* 111, 124046 (2025). DOI: 10.1103/h4nh-nl5s. arXiv: 2503.10746 [gr-qc].
I had the leading role in developing the analysis tools for the paper, as well as in the writing.

[20] A. Colombo, O. S. Salafia, et al. (incl. **F. Iacovelli**). "Multi-messenger observations in the Einstein Telescope era: binary neutron star and black hole - neutron star mergers". In *Astron. Astrophys.* 704 (2025) A260. DOI: 10.1051/0004-6361/202554326. arXiv: 2503.00116 [astro-ph.HE].
I took care of the gravitational wave modeling aspects of the paper, writing the relative part and producing the results of the gravitational wave observational prospects.

[19] M. Maggiore, **F. Iacovelli**, E. Belgacem, M. Mancarella, N. Muttoni. "Comparison of global networks of third-generation gravitational-wave detectors". In *Class. Quantum Grav.* 42 215004

(2025). DOI: 10.1088/1361-6382/ae110b. arXiv: 2411.05754 [gr-qc].
I produced most of the numerical results and plots.

[18] E. Belgacem, **F. Iacovelli**, M. Maggiore, M. Mancarella, N. Muttoni. "The spectral density of astrophysical stochastic backgrounds". In *JCAP* 04 (2025) 032. DOI: 10.1088/1475-7516/2025/04/032. arXiv: 2411.04028 [gr-qc].
I contributed to the theoretical development of the paper, the writing and produced the numerical results.

[17] V. De Renzis, **F. Iacovelli**, D. Gerosa, M. Mancarella, C. Pacilio. "Forecasting the population properties of merging black holes". In *Phys. Rev. D* 111, 044048 (2025). DOI: 10.1103/PhysRevD.111.044048. arXiv: 2410.17325 [astro-ph.HE].
I had a major role in the theoretical development and writing of the paper, and in developing the associated code.

[16] M. Mancarella, **F. Iacovelli**, S. Foffa, N. Muttoni, M. Maggiore. "Accurate standard siren cosmology with joint gravitational-wave and γ-ray burst observations". In *Phys. Rev. Lett.* 133, 261001 (2025). DOI: 10.1103/PhysRevLett.133.261001. arXiv: 2405.02286 [astro-ph.HE].
I took part in the development and implementation of the new GW approximant and the inclusion of noise scatter in the measurement, as well as in the reviewing of the hierarchical bayesian likelihood and writing.

[15] **F. Iacovelli**, E. Belgacem, M. Maggiore, M. Mancarella, N. Muttoni. "Combining underground and on-surface third-generation gravitational-wave interferometers". In *JCAP* 10 (2024) 085. DOI: 10.1088/1475-7516/2024/10/085. arXiv: 2408.14946 [gr-qc].

[14] F. Santoliquido, U. Dupletsa, et al. (incl. **F. Iacovelli**). "Classifying binary black holes from Population III stars with the Einstein Telescope: a machine-learning approach". In *Astron. Astrophys.* 690 (2024) A362. DOI: 10.1051/0004-6361/202450381. arXiv: 2404.10048 [astro-ph.HE].
I participated in the writing of some parts of the work as well as in discussions regarding the set-up of the analysis and helped perform some initial checks among GWFISH and GWFAST.

[13] P. Ajith, et al. (incl. **F. Iacovelli**). "The Lunar Gravitational-wave Antenna: Mission Studies and Science Case". In *JCAP* 01 (2025) 108. DOI: 10.1088/1475-7516/2025/01/108. arXiv: 2404.09181 [gr-qc].
I coordinated the work and writing of the GW cosmology part of the paper and contributed to other sections, including forecasts for localization and multiband capabilities with LGWA.

[12] A. Colombo, R. Duqué, et al. (incl. **F. Iacovelli**). "Multi-messenger prospects for black hole – neutron star mergers in the O4 and O5 runs". In *Astron. Astrophys.* 686 (2024) A265. DOI: 10.1051/0004-6361/202348384. arXiv: 2310.16894 [astro-ph.HE].
I took care of the gravitational wave modeling aspects of the paper, writing the relative part and producing the results of the gravitational wave observational prospects.

[11] N. Borghi, M. Mancarella, M. Moresco, M. Tagliazucchi, **F. Iacovelli**, A. Cimatti, M. Maggiore. "Cosmology and Astrophysics with Standard Sirens and Galaxy Catalogs in View of Future Gravitational Wave Observations". In *Astrophys. J.* 964 (2024) 2, 191. DOI: 10.3847/1538-4357/ad20eb. arXiv: 2312.05302 [astro-ph.CO].
I took part in the discussions regarding the theoretical and methodological aspects of the paper as well as in the writing.

[10] **F. Iacovelli**, M. Mancarella, et al. "Nuclear physics constraints from binary neutron star mergers in the Einstein Telescope era". In *Phys. Rev. D* . 108, 122006 (2023). DOI: 10.1103/PhysRevD.108.122006. arXiv: 2308.12378 [gr-qc].

[9] G. Franciolini, **F. Iacovelli**, M. Mancarella, M. Maggiore, P. Pani and A. Riotto. "Searching for Primordial Black Holes with the Einstein Telescope: impact of design and systematics". In *Phys. Rev. D* . 108, 043506 (2023). DOI: 10.1103/PhysRevD.108.043506. arXiv: 2304.03160 [gr-qc].
My contribution can be considered as first author. I had a major role in the theoretical development and writing of the paper, and I also produced many of the reported results.

[8] M. Branchesi, M. Maggiore, et al. (incl. **F. Iacovelli**). "Science with the Einstein Telescope: a comparison of different designs". In *JCAP* 07 (2023) 068. DOI: 10.1088/1475-7516/2023/07/068. arXiv: 2303.15923 [gr-qc].
I took part in the development of various sections of the work, producing in particular most of

the results in the "Coalescence of compact binaries" section, and participating in the "Nuclear physics", "Population studies" and "Cosmology" sections.

[7] M. Mancarella, **F. Iacovelli**, D. Gerosa. "Inferring, not just detecting: metrics for high-redshift sources observed with third-generation gravitational-wave detectors". In *Phys. Rev. D* . 107, L101302 (2023). DOI: 10.1103/PhysRevD.107.L101302. arXiv: 2303.16323 [gr-qc].
I participated in the theoretical development and writing of the paper, also producing part of the reported results (in particular the 'inference horizon').

[6] M. Kole, **F. Iacovelli**, M. Mancarella, N. Produit. "Adding Gamma-ray Polarimetry to the Multi-Messenger Era". In *Astron. Astrophys.* 669 (2023) A77. DOI: 10.1051/0004-6361/202245205. arXiv: 2211.12403 [astro-ph.HE].
I produced the results and plots of the gravitational wave part, also participating in the theoretical development of the paper.

[5] **F. Iacovelli**, M. Mancarella, S. Foffa, M. Maggiore. "Forecasting the detection capabilities of third-generation gravitational-wave detectors using GWFAST". In *Astrophys. J.* 941 (2022) 2, 208. DOI: 10.3847/1538-4357/ac9cd4. arXiv: 2207.02771 [gr-qc].

[4] **F. Iacovelli**, M. Mancarella, S. Foffa, M. Maggiore. "GWFAST: A Fisher Information Matrix Python Code for Third-generation Gravitational-wave Detectors". In *Astrophys. J. Supp.* 263 (2022) 1, 2. DOI: 10.3847/1538-4365/ac9129. arXiv: 2207.06910 [astro-ph.IM].

[3] A. Finke, S. Foffa, **F. Iacovelli**, M. Maggiore, and M. Mancarella. "Modified gravitational wave propagation and the binary neutron star mass function". In: *Phys. Dark Univ.* 36 (2022) 100994. DOI: 10.1016/j.dark.2022.100994. arXiv: 2108.04065 [gr-qc].
I produced most of the results results, figures, and code, developing also a useful data visualization technique.

[2] A. Finke, S. Foffa, **F. Iacovelli**, M. Maggiore, and M. Mancarella. "Probing modified gravitational wave propagation with strongly lensed coalescing binaries". In: *Phys. Rev. D* 104.8 (2021), p. 084057. DOI: 10.1103/PhysRevD.104.084057. arXiv: 2107.05046 [gr-qc].
I produced most of the results results and figures within this paper, wrote the necessary code, and also contributed to the development of the theoretical framework.

[1] A. Finke, S. Foffa, **F. Iacovelli**, M. Maggiore, and M. Mancarella. "Cosmology with LIGO/Virgo dark sirens: Hubble parameter and modified gravitational wave propagation". In: *JCAP* 08 (2021), p. 026. DOI: 10.1088/1475-7516/2021/08/026. arXiv: 2101.12660 [astro-ph.CO].
This work is part of my Master's thesis, and I contributed writing parts of the accompanying code, studying in depth and handling the problem of the completeness of galaxy catalogs, as well as producing most of the figures and results.

Papers under review

[4] **F. Iacovelli**, L. Reali, E. Berti, A. Corsi, B. S. Sathyaprakash, D. Wadekar. "Not too close! Evaluating the impact of the baseline on the localization of binary black holes by next-generation gravitational-wave detectors". 2026. arXiv: 2604.11871 [gr-qc].

[3] S. Yi, **F. Iacovelli**, E. Berti, R. S. Chandramouli, S. Marsat, D. Wadekar, N. Yunes. "Systematic biases in parameter estimation on LISA binaries. II. The effect of excluding higher harmonics for spin-aligned, high-mass binaries". 2026. arXiv: 2602.09088 [gr-qc].
I contributed to the conceptual development of the paper, the writing, and the improvement and validation of the likelihood optimization tool.

[2] **F. Iacovelli**, J. Tissino, J. Harms, E. Berti. "Gravitational-wave parameter estimation to the Moon and back: massive binaries and the case of GW231123". 2025. arXiv: 2512.09978 [gr-qc].

[1] A. Leviyev, **F. Iacovelli**, A. Zimmerman. "Efficient Bayesian Sampling with Langevin Birth-Death Dynamics". 2025. arXiv: 2509.01942 [stat.AP].
I contributed to the production of the results related to the gravitational-wave parameter estimation, text writing, and discussion on the sampling algorithm.

Public softwares

[3] V. De Renzis, **F. Iacovelli**. GWFAST-population: a Fisher information matrix package for population analyses at GW detector networks. Git: https://github.com/CosmoStatGW/gwfast/tree/master/gwfast/population.

[2] **F. Iacovelli**, M. Mancarella. GWFAST: a Fisher information matrix Python package for GW detector networks. ascl: 2212.001. Git: https://github.com/CosmoStatGW/gwfast.

[1] **F. Iacovelli**. WF4Py: Gravitational waves waveform models in pure Python language. ascl: 2301.003 Git: https://github.com/CosmoStatGW/WF4Py.

Proceedings

[5] **F. Iacovelli**, M. Mancarella., S. Foffa, M. Maggiore. "GWFAST: A tool to explore the capabilities of Third-generation Gravitational-wave Interferometers". In *Proceedings, PUMA22*. 2022. *Genova University Press* "Probing the Universe with Multimessenger Astronomy" p.68. e-ISBN: 978-88-3618-218-3.

[4] M. Mancarella, N. Borghi, S. Foffa, E. Genoud-Prachex, **F. Iacovelli**, M. Maggiore, M. Moresco, M. Schulz. "Gravitational-wave cosmology with dark sirens: state of the art and perspectives for 3G detectors". In *PoS ICHEP2022 127*. 2022. arXiv: 2211.15512 [gr-qc].

[3] **F. Iacovelli**, A. Finke, S. Foffa, M. Maggiore, M. Mancarella. "Modified gravitational wave propagation: information from strongly lensed binaries and the BNS mass function". In *Proceedings, 56th Rencontres de Moriond on Gravitation*. 2022. arXiv: 2203.09237 [gr-qc].

[2] M. Mancarella, A. Finke, S. Foffa, E. Genoud-Prachex, **F. Iacovelli**, M. Maggiore. "Cosmology and modified gravity with dark sirens from GWTC-3". In *Proceedings, 56th Rencontres de Moriond on Gravitation*. 2022. arXiv: 2203.09238 [gr-qc].

[1] A. Finke, S. Foffa, **F. Iacovelli**, M. Maggiore, and M. Mancarella. "Constraining the Hubble constant and modified GW propagation with LIGO/Virgo dark sirens". In: *Proceedings, 55th Rencontres de Moriond on Cosmology*. 2021.

Other

[3] **F. Iacovelli**. "What does the Universe sound like?". 2026. April Fools'. arXiv: 2603.29996 [gr-qc].

[2] E. Berti, et al. (incl. **F. Iacovelli**). "deci-Hz Gravitational Wave Observations on the Moon and Beyond". 2026. arXiv: 2602.05923 [gr-qc].

[1] **F. Iacovelli**, M. Maggiore. "Gravitational-Wave Observations and Primordial Black Holes". Chapter 15 in *Primordial Black Holes*, 2025. DOI: 10.1007/978-981-97-8887-3_15. arXiv: 2407.21442 [gr-qc].

Talks

Invited seminars at international research institutes: **7** invited presentations in international institutes, **1** invited presentation at collaboration call.

Invited contributions at conferences and workshops: **6** invited presentations (1 upcoming), **2** invited presentations on behalf of the Einstein Telescope Collaboration, **2** invitations as moderator in workshop discussion panels.

Contributed talks at conferences and workshops: **14** contributed presentations.

Invited contributions

[18] **APS Global Physics Summit**, Denver, US. March 15-20, 2026. Title: *Gravitational waves to the Moon: prospects and science case*.

[17] **Sapienza University** (Italy). September 24th, 2025. Title: *Forecasting the population properties of merging black holes*.

[16] **LGWA workshop**, University of Camerino, San Benedetto del Tronto, Italy. September 15–19, 2025. Title: *Cosmology in the dHz: opportunities from single sources and stochastic backgrounds*.

[15] **deciHz Gravitational Wave Observations on the Moon and Beyond**, Johns Hopkins University, Baltimore, US. September 1–3, 2025. *Moderation* in the panels ``*deciHz gravitational-wave science: astrophysical and cosmological sources*" and ``*deciHz experiments and science*".

[14] **Fundamental Physics Across the Gravitational Wave Spectrum**, Kavli Institute for Cosmological Physics at the University of Chicago, Chicago, US. August 20–22, 2025. Title: *Characterizing the detectability of cosmological backgrounds at third-generation detectors*.

[13] **10th Physics and Astrophysics at the eXtreme Workshop and 3rd Cosmic Explorer Symposium**, University of Illinois at Urbana-Champaign, Urbana, US. June 30–July 3, 2025. *Moderation* in the panel "*Data analysis in the 100,000+ events per year era*".

[12] **Aix Marseille Université** (France). June 6th, 2025. Title: *Characterizing the detectability of cosmological backgrounds at 3G detectors*.

[11] **Padova University** (Italy). Cosmology Journal Club, April 9th, 2025. Title: *Cosmology with third-generation gravitational-wave detectors: from individual sources to stochastic backgrounds*.

[10] **Princeton University – Institute for Advanced Study** (US). Cosmology Lunch Seminar, February 3rd, 2025. Title: *Cosmology with third-generation gravitational-wave detectors: from individual sources to stochastic backgrounds*.

[9] **1st GSSI-JHU MAECI meeting on Gravitational Wave Astrophysics**, L'Aquila, Italy. December 16–18, 2024. Title: *Challenges for population analyses at 3G*.

[8] **Johns Hopkins University** (US). CAS Wine and Cheese Seminars, October 28th, 2024. Title: *Forecasting the Capabilities of Third–Generation Gravitational–Wave Detectors: from One to Many*.

[7] **IV Gravi-Gamma-Nu Workshop**, Gran Sasso Science Institute, L'Aquila, Italy. October 4–6, 2023. Title: *Einstein Telescope: science objectives and designs*. Talk on behalf of the Einstein Telescope Collaboration.

[6] **XXV SIGRAV Conference**, Scuola Internazionale Superiore di Studi Avanzati, Trieste, Italy. September 4–8, 2023. Title: *Detection and measurement prospects at the Einstein Telescope: forecasts with GWFAST*.

[5] **XIII Einstein Telescope Symposium**, Cagliari, Italy. May 8–12, 2023. Title: *GWFAST: a Fisher-matrix Python code for third-generation gravitational-wave detectors*.

[4] **57th Rencontres de Moriond - Gravitation**, March 2023, La Thuile (IT). Talk title *Einstein Telescope: science objectives and current status*. Talk on behalf of the Einstein Telescope Collaboration.

[3] **University of Milan-Bicocca** (Italy). AstroBicocca Seminars, January 26th, 2023. Title: *Forecasting the detection capabilities of third-generation gravitational-wave detectors using GWFAST*.

[2] **COSI Collaboration GRB working group** (online). October 20th, 2023. Title: *Adding Gamma-ray Polarimetry to the Multi-Messenger Era: gravitational–wave detectors prospects*.

[1] **Gran Sasso Science Institute** (Italy). September 22nd, 2022. Title: *Forecasting the detection capabilities of third-generation gravitational-wave detectors using GWFAST*.

Contributed talks

[14] **XXVI SIGRAV Conference**, University of Milano-Bicocca, Milan, Italy. September 8–12, 2025. Title: *Characterizing the detectability of cosmological backgrounds at third-generation detectors*.

[13] **2025 HotSci Series**, Johns Hopkins University and Space Telescope Science Institute, Baltimore, US. July 9th 2025. Title: *Prospects of bright-siren cosmology with third-generation gravitational-wave detectors*.

[12] **XV Einstein Telescope Symposium**, University of Bologna, Bologna, Italy. May 26–30, 2025. Title: *Tainted Love: Systematic biases from ignoring environmental tidal effects in gravitational wave observations*.

[11] **59th Rencontres de Moriond - Gravitation**, La Thuile, Italy. March 30–April 6, 2025. Title: *Accurate Standard Siren Cosmology with Joint Gravitational-Wave and γ-Ray Burst Observations*.

[10] **XIV Einstein Telescope Symposium**, Maastricht, Netherlands. May 6–10, 2024. Title: *Extending the Fisher matrix formalism towards the edges*.

[9] **2nd Cosmic Explorer Symposium**, online. April 23–25, 2024. Title: *Accurate Standard Siren Cosmology with Joint Gravitational-Wave and γ-Ray Burst Observations*.

[8] **15th Edoardo Amaldi Conference on Gravitational Waves**, online. July 17–21, 2023. Title: *Forecasting the detection capabilities of third–generation gravitational–wave detectors using GWFAST*.

[7] **Third EuCAPT Annual Symposium**, CERN, Geneva, Switzerland. May 31–June 2, 2023. Title: *Searching for Primordial Black Holes with the Einstein Telescope*.

[6] **XIII Einstein Telescope Symposium**, Cagliari, Italy. May 8–12, 2023. Title: *Nuclear Physics with ET: a comparison of different designs*.

[5] **PUMA22 Conference**, Sestri Levante, Italy. September 26–30, 2022. Title: *Forecasting the detection capabilities of third–generation gravitational–wave detectors using GWFAST*.

[4] **56th Rencontres de Moriond - Gravitation**, La Thuile, Italy. January 30– February 6, 2022. Title: *Modified gravitational wave propagation: information from strongly lensed binaries and the BNS mass function*.

[3] **XXIV SIGRAV Conference**, Urbino, Italy. September 7–9, 2021. Title: *Cosmology with LIGO/Virgo dark sirens and galaxy catalogs*.

[2] **14th Edoardo Amaldi Conference on Gravitational Waves**, online. July 19–23, 2021. Title: *Cosmology with LIGO/Virgo dark sirens and galaxy catalogs*.

[1] **First EuCAPT Annual Symposium**, online. May 5–7, 2021. Title: *Cosmology with LIGO/Virgo dark sirens and galaxy catalogs*.

Other items of interest

Scientific events organization

- Einstein Telescope Early Career Researchers workshop, February 18-20, 2026.
- deci-Hz gravitational wave observations on the Moon and beyond, September 1-3, 2025.
- Hackathon at the XIII Einstein Telescope Symposium, May 2023.

Reviewer: Physical Review D, Physical Review Letters, International Journal of Modern Physics D, Journal of Cosmology and Astroparticle Physics, External evaluator for ERC Starting Grant.

Fellowships, funding, grants:

2024–current **JHU Miller Postdoctoral Fellowship**

2022–2023 **Istituto Svizzero Milano Calling Fellowship**

2019–2020 **Excellence program at Sapienza University**

2018–2019 **Collaboration Scholarship at Sapienza University**